Semester	T.E. Semester VI – Computer Engineering
Subject	E- waste
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E-Waste Management – Solution

Understanding the Problem
E-waste is growing rapidly mainly because of the **increasing consumption of electronic products**, the **digital revolution**, and continuous **technological innovation**. People and industries frequently adopt newer electronic devices, which causes older devices and components to be discarded more quickly. The paper reports that **62 billion kg of e-waste was generated globally in 2022, while only 22.3% was recycled**.

The literature review specifically connects the huge volume of e-waste with the growing consumption of electronic goods due to the digital revolution and advances in science and technology.

E-waste is a serious environmental and health problem because discarded electronic equipment can contain **hazardous or toxic materials**. If it is dumped or recycled improperly, these substances can contaminate **air, water and soil**, creating risks to human health as well as environmental sustainability. The problem is particularly serious in India because the paper describes much e-waste disposal as informal and poorly structured.

In simple words:

We are buying and replacing electronics faster, so more electronic products are becoming waste. Since e-waste contains both hazardous substances and recoverable materials, improper disposal causes pollution, health risks and loss of useful resources.

Solution Approach — AHP and ISM

AHP (Analytic Hierarchy Process): The main purpose of AHP was to **prioritize and rank the identified e-waste management challenges**. Experts compared the barriers, and the researchers calculated a weight for each challenge so they could determine which challenges were more important. AHP was also used to check the consistency of the experts' opinions.

ISM (Interpretive Structural Modeling): The main purpose of ISM was to **identify relationships among the challenges and arrange them into different levels**. Instead of simply saying which challenge is more important, ISM shows how one barrier can influence another and creates a structured model showing which barriers need attention from the more basic/root level toward more complex, long-term barriers.

Easy difference to remember
AHP = "Which challenge is more important?" → Ranking and priority
ISM = "How are the challenges connected?" → Relationships and levels
So, **AHP gives the priority/weight of the barriers, while ISM gives their structural relationship and order**.